

NAME: _____

PERIOD: _____

Superpowers: Frequency Table + Bar Chart

AP Statistics · Unit 1 Poster · Group sheet 1 of 5

[STAR] YOUR POSTER SHOWS a frequency table AND a bar chart of the same data

Data (Lesson 1.3). A random sample of 50 high school students was asked which superpower they would most like to have.

Superpower	Fly	Freeze time	Invisibility	Super strength	Telepathy
Students	9	15	7	3	16

[WARN] THE CLAIM YOU ARE JUDGING

A streaming service is pitching a show called *Freeze Time*: “Freeze time is the superpower teens want most, and it is twice as popular as flying.” A second writer says the show should be about telepathy.

DOK 1 (a) Name the individuals and the variable. Is the variable categorical or quantitative? Why does that call for a bar chart instead of a dotplot or histogram?

DOK 2 (b) Check the pitch’s two claims with the table. Is freeze time the *most* chosen? Is it *twice* as popular as flying? Show the numbers.

DOK 2 (c) Write one claim about the two most popular superpowers that the data *does* support. Include a number.

DOK 3 (d) ★ **For the poster:** in two or three sentences, say what the data supports, what it does not, and the honest claim. Name the context and use at least two numbers.

[TARGET] POSTER CHECKLIST · Data · Graph · Sentence

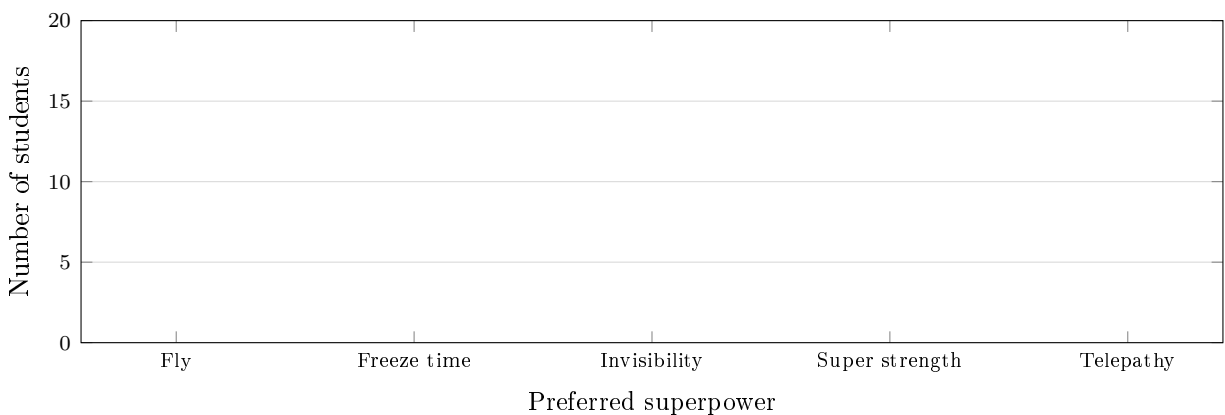
- ☐ **Data:** the frequency table with a total row (adds to 50)
- ☐ **Graph:** bar chart: title, both axes labeled, bars separated, count scale that fits 16
- ☐ **Sentence:** individuals, variable, and “categorical” in one sentence
- ☐ **Sentence:** “most” and “twice” each judged with numbers
- ☐ **Sentence:** one honest claim, in context, with a number

All five = E. Three or four = P. Fewer = I.

[PUZZLE] ROLES · solve (a)–(d) together first, then split

Data Keeper: table, total, the “most” / “twice” arithmetic. **Grapher:** the bar chart. **Writer:** the three sentences.

Sketch the graph here first. Bars separated, one per superpower.



Frame (optional): In a sample of _____, the most chosen superpower was _____, with _____ students (____). Freeze time was chosen about _____ as often as flying, so the pitch is _____.
how many who which count percent how many times supported or not

Word bank: telepathy · freeze time · 1.7 times · twice · 32% · 30% · 62% · not supported · supported
Some words are not needed.

Draft the poster sentences here. Every sentence names the context and uses a number from the graph.

NAME: _____

PERIOD: _____

Spirit Day: Relative Frequency Table + Bar Chart

AP Statistics · Unit 1 Poster · Group sheet 2 of 5

[STAR] YOUR POSTER SHOWS a relative frequency table for both grades **AND** a side-by-side relative frequency bar chart

Data (Lesson 1.4). A teacher surveyed two grades about their favorite spirit day theme.

Theme	9th grade (30 students)	10th grade (20 students)
Pajama Day	12	5
Jersey Day	6	8
Twin Day	9	3
Costume Day	3	4

[WARN] THE CLAIM YOU ARE JUDGING

Student council picked Jersey Day for the whole school: *“Jersey Day got the most 10th-grade votes, and 9th grade was close anyway: 6 versus 8.”*

DOK 1 (a) Turn every count into a percent of its own grade (divide by 30 or by 20). Check that each column totals 100%.

DOK 2 (b) Which theme does each grade actually prefer? Is “6 versus 8 is close” still true in percents?

DOK 2 (c) In one or two sentences: why would a chart of raw counts have misled the council, and what does dividing by the group size fix?

DOK 3 (d) ★ **For the poster:** the grades disagree. Recommend one schoolwide theme and defend it with percents in two or three sentences. Any theme is fine if the percents back it.

[TARGET] POSTER CHECKLIST · Data · Graph · Sentence

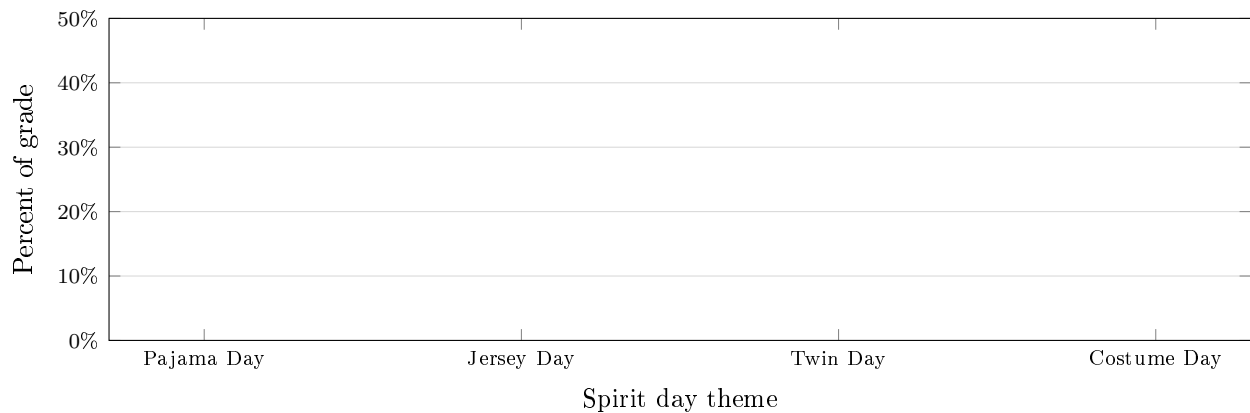
- ☐ **Data:** relative frequency table, eight percents, both columns total 100%, two divisions shown
- ☐ **Graph:** side-by-side relative frequency bar chart: title, percent axis, key for the two grades
- ☐ **Sentence:** each grade's favorite, with its percent
- ☐ **Sentence:** "6 versus 8 is close" rejected with 20% vs 40%
- ☐ **Sentence:** the recommendation defended with percents, plus why counts mislead when groups differ in size

All five = E. Three or four = P. Fewer = I.

[PUZZLE] ROLES · solve (a)–(d) together first, then split

Data Keeper: the eight percents and column totals. **Grapher:** the side-by-side chart with key. **Writer:** favorites, the recommendation, the "why counts mislead" sentence.

Sketch the graph here first. Two bars per theme, one shade per grade, with a key.



Frame (optional): Jersey Day was chosen by _____ of 9th graders but _____ of 10th graders, so the
grades were _____. The council compared _____ from groups of _____ and _____, so
_____.
_____.

percent *percent* *close or not* *counts or percents* *size* *size* *what went wrong*

Word bank: 20% · 40% · 25% · not close · close · counts · percents · 30 · 20 · the bigger grade looked more enthusiastic · the smaller grade looked more enthusiastic Some words are not needed.

Draft the poster sentences here. Every sentence names the context and uses a number from the graph.

Late Bus: Dotplot

AP Statistics · Unit 1 Poster · Group sheet 3 of 5

[STAR] YOUR POSTER SHOWS a dotplot: every value as a dot on a number line with a scale

Data (Lesson 1.6). A school recorded how many minutes 11 students waited for a late bus after school.

Student	1	2	3	4	5	6	7	8	9	10	11
Minutes waiting	1	2	3	3	4	4	4	5	5	6	18

[WARN] THE CLAIM YOU ARE JUDGING

The transportation office reports: *“Students wait an average of 5 minutes for the late bus, so there is no problem.”*

DOK 1 (a) Describe the distribution in context, one labeled line each: shape, center, variability, unusual features. (Sketch the dotplot on page 2 first if it helps.)

DOK 2 (b) Find the five-number summary and the IQR. Is 18 an outlier by the $1.5 \times \text{IQR}$ rule? Show the fence.

DOK 2 (c) The mean is $55 \div 11 = 5.0$ minutes and the median is 4. Which dot opens that gap? Which measure describes a typical wait, and why?

DOK 3 (d) ★ **For the poster:** explain why “average of 5, no problem” hides the real problem, then rewrite the office’s report honestly in one or two sentences, with numbers.

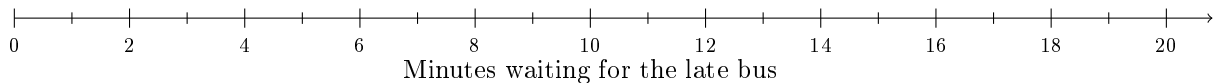
[TARGET] POSTER CHECKLIST · Data · Graph · Sentence

- ☐ **Data:** the 11 values, the five-number summary, the IQR
 - ☐ **Graph:** dotplot: 11 dots, number line 0 to 20 with even ticks, axis label with units, title
 - ☐ **Sentence:** shape, center, variability, unusual features, each labeled and in context
 - ☐ **Sentence:** the outlier check: Q1 3, Q3 5, IQR 2, fence $5 + 3 = 8$, so 18 is an outlier
 - ☐ **Sentence:** the honest rewrite, with mean 5.0 vs median 4 explained from the dotplot
- All five = E. Three or four = P. Fewer = I.

[PUZZLE] ROLES · solve (a)–(d) together first, then split

Data Keeper: five-number summary, IQR, fence, mean. **Grapher:** the dotplot. **Writer:** the four description lines and the honest rewrite.

Sketch the graph here first. One dot per student; stack repeats.



Frame (optional): The distribution of _____ is _____, with a typical wait of about _____ minutes. Waits ranged from _____ to _____ minutes; _____ is an outlier because it is above the fence of _____. The office's "average of 5" misleads because _____.

variable in context *shape*
number *min* *max* *which value* *fence* *reason*

Word bank: minutes waiting for the late bus · skewed right · skewed left · symmetric · 4 · 5 · 1 · 18 · 8 · one long wait pulled the mean up · the median is not resistant Some words are not needed.

Draft the poster sentences here. Every sentence names the context and uses a number from the graph.

NAME: _____

PERIOD: _____

Resting Heart Rate: Stem-and-Leaf Plot + Histogram

AP Statistics · Unit 1 Poster · Group sheet 4 of 5

[STAR] YOUR POSTER SHOWS a stem-and-leaf plot AND a histogram (bins of 5 bpm) of the same 25 values, side by side

Data (Lesson 1.2 spring-sport context). Resting heart rate, in beats per minute (bpm), for 25 students who play a spring sport, already in order:

42, 46, 50, 54, 57, 58, 59, 60, 60, 61, 62, 62, 62, 63, 64, 64, 65, 65, 66, 66, 67, 68, 68, 69, 70

[WARN] THE CLAIM YOU ARE JUDGING

The coach says: *“Resting heart rate is skewed, so the mean is pulled up. I will use the mean as the typical athlete, and anyone below 50 bpm has to see the trainer.”*

DOK 1 (a) Make the stem-and-leaf plot (stems 4 to 7, leaves in order, a key such as $6 \mid 2 = 62$ bpm). Tally the bins 40–44, 45–49, 50–54, 55–59, 60–64, 65–69, 70–74. Use the sketch boxes on page 2.

DOK 2 (b) Shape: which way does the tail point? Which way does that pull the mean, above or below the median? Check with the actual mean (the 25 values sum to 1528) and the median.

DOK 2 (c) Find Q1, Q3, the IQR and the lower fence. Which athletes are outliers by the $1.5 \times \text{IQR}$ rule? Does “below 50” match?

DOK 3 (d) ★ **For the poster:** in two or three sentences, which center should the coach report and why; and the trainer wants each athlete’s exact number, so which display do you hand her and what does the other one hide?

[TARGET] POSTER CHECKLIST · Data · Graph · Sentence

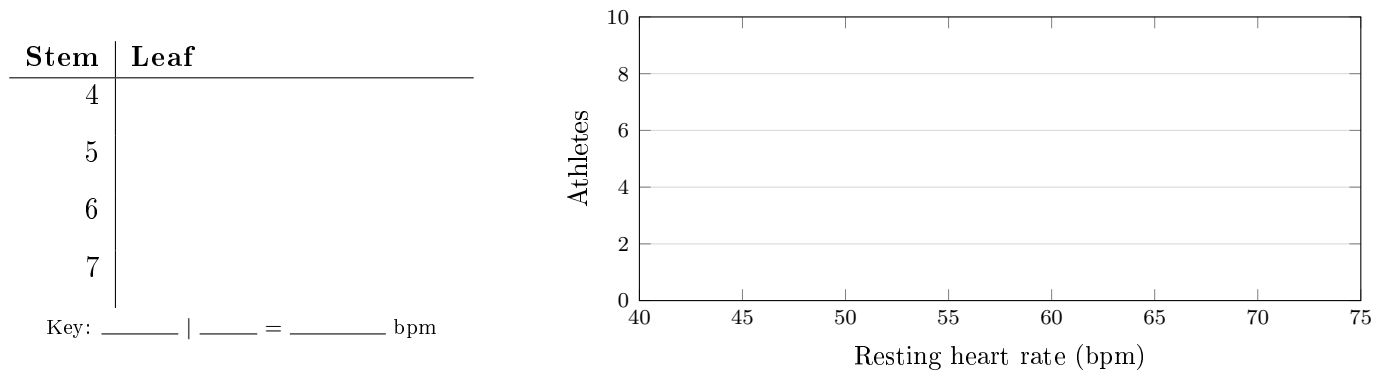
- ☐ **Data:** the 25 values; mean 61.1, median 62, Q1 58.5, Q3 66, IQR 7.5, lower fence 47.25
- ☐ **Graph:** stem-and-leaf plot, stems 4–7, leaves in order, with a key
- ☐ **Graph:** histogram, bins 40–44 to 70–74, bars touching, axes labeled with units, title
- ☐ **Sentence:** “skewed left” with the tail direction, and mean vs median confirming it
- ☐ **Sentence:** which center to report and why; which display shows every athlete and what the histogram hides

All five = E. Three or four = P. Fewer = I.

[PUZZLE] ROLES · solve (a)–(d) together first, then split

Data Keeper: mean, median, quartiles, fence, outliers. **Grapher:** both displays, side by side. **Writer:** shape and direction, the center choice, the trainer sentence.

Sketch the graph here first. Stem-and-leaf on the left, histogram on the right.



Frame (optional): The distribution of _____ is skewed _____, so the mean of _____ is _____ the median of _____. The coach should report the _____ because _____. The trainer gets the _____ because _____.

variable in context
direction
value

above or below
value
which
reason

which display
what the other hides

Word bank: resting heart rate in bpm · left · right · 61.1 · 62 · below · above · median · mean · it resists the two low outliers · it follows the two low outliers · stem-and-leaf plot · histogram · bins hide the exact values · Some words are not needed.

Draft the poster sentences here. Every sentence names the context and uses a number from the graph.

Study Halls: Parallel Box Plots

AP Statistics · Unit 1 Poster · Group sheet 5 of 5

[STAR] YOUR POSTER SHOWS two box plots on **ONE** shared number line, the outlier as a separate point

Data (Lesson 1.9). Nightly homework time, in minutes, for students in two study halls.

	Minimum	Q1	Median	Q3	Largest non-outlier	Outliers
Study Hall A	30	45	60	90	110	one at 150
Study Hall B	45	54	60	66	75 (maximum)	none

[WARN] THE CLAIMS YOU ARE JUDGING

The principal: *“Both study halls have a median of 60 minutes, so homework time is the same in both.”*

A parent: *“What percent of Study Hall A students spent more than 90 minutes?”*

DOK 1 (a) Sketch both box plots on the number line on page 2. Then find each study hall’s IQR and range.

DOK 2 (b) Compare the two: shape, center, variability, unusual features. Every line needs a comparative word (greater, less, similar, the same) and the context (homework minutes).

DOK 2 (c) Answer the parent. How does a box plot let you say that without the individual data?

DOK 3 (d) ★ **For the poster:** is the principal right? Use two features besides the median. Then: which study hall has the most predictable homework time, and which box-plot feature proves it? Two or three sentences.

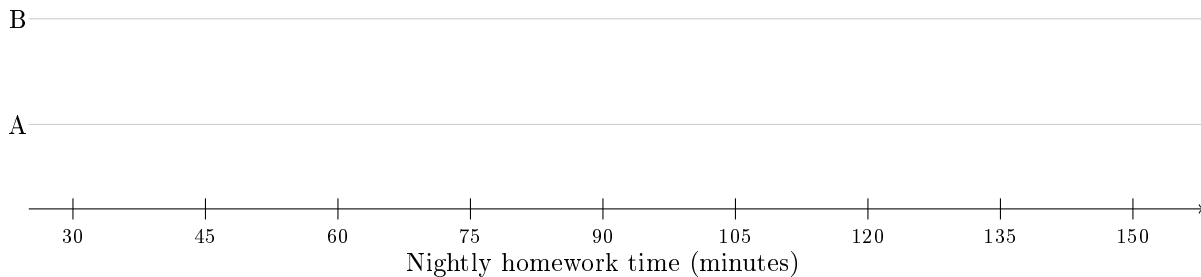
[TARGET] POSTER CHECKLIST · Data · Graph · Sentence

- ☐ **Data:** both five-number summaries, both IQRs and ranges
 - ☐ **Graph:** two box plots on one scale (30 to 150), labeled A and B, axis label with units, 150 as a separate point, each section labeled 25%
 - ☐ **Sentence:** four comparison lines (shape, center, variability, unusual), each with a comparative word, in context
 - ☐ **Sentence:** the parent answered (about 25%) with the “each section is 25%” rule
 - ☐ **Sentence:** the principal judged with numbers (IQR 45 vs 12, range 120 vs 30, outlier vs none) and the predictability pick tied to the narrower box
- All five = E. Three or four = P. Fewer = I.

[PUZZLE] ROLES · solve (a)–(d) together first, then split

Data Keeper: IQRs, ranges, the 25% labels. **Grapher:** both plots on one scale, outlier as a point.
Writer: four comparison lines, the parent’s answer, the pick.

Sketch the graph here first. Study Hall A on the top row, B on the bottom, same number line.



Frame (optional): The median homework time is _____ for both study halls at _____ minutes, but
*comparative**value*
Study Hall A’s times are _____ spread out (IQR _____ vs _____ minutes) and A has _____.
*comparative**value**value**unusual feature*
About _____ of A spent more than 90 minutes because _____.
*percent**box plot rule*

Word bank: the same · greater · less · far more · 60 · 45 · 12 · 25% · 50% · 75% · one high outlier at 150 · no outliers · each section of a box plot holds about a quarter of the data Some words are not needed.

Draft the poster sentences here. Every sentence names the context and uses a number from the graph.